

Annotation Cascading: Hierarchical Model Routing, Topical Authority, and Inter-Page Context Propagation in Large-Scale Web Content Classification

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Abstract

Web-scale content annotation — the process by which search engines and AI systems classify, tag, and evaluate web content for downstream retrieval and generation — operates under extreme computational constraints. This paper makes three contributions. (1) We present *Annotation Cascading*: a framework modelling how content routes through a hierarchy of increasingly specialised language models, from site-level broad classification to chunk-level entity resolution, in which the LLM serves as the computational fallback rather than the primary annotation mechanism. We identify *confident misannotation* — domain routing error compounded by control-plane failure — as a structurally distinct failure mode. (2) We reframe topical authority — a concept widely discussed in the search engine optimisation literature but never formally defined in information retrieval — as measurable annotation routing efficiency across three dimensions (Closeness, Strength, Longevity), grounded in Knowledge Graph entity-topic-URL associations, with a formal cost model and operationalisable metrics. (3) We propose the *Navigation Memory hypothesis*: that annotation context propagates between pages during a crawl session, introducing the distinction between Carry Anchors (context-preserving links) and Break Anchors (context-resetting links), with two experimental designs. We present a predictions-to-proxies mapping and testable measurement protocols for empirical validation.

Keywords: Annotation Cascading, topical authority, Navigation Memory, hierarchical model routing, SLM routing, web content classification, Assistive Agent Optimization, AAO, entity annotation, Knowledge Graph, multi-stage classification, adaptive inference, confident misannotation

1. Introduction

When a search engine crawler processes a web page, it must answer several questions simultaneously: What is this content about? Which entities appear in it? How does it relate to established concepts in the system’s knowledge representations? What confidence should the system assign to these classifications? The answers — collectively, the annotation of the content

— determine whether and how the content enters knowledge graphs, search indexes, and language model training corpora.

The computational economics are severe. Major search engines discover billions of URLs daily. Annotating each page with a state-of-the-art large language model would be prohibitively expensive. The engineering community has acknowledged that smaller, domain-specific models handle annotation where possible (Nayak, 2022; Microsoft Bing, 2024), but the information retrieval literature has not formalised the routing mechanics, failure modes, or downstream consequences of this architectural decision.

This paper addresses that gap. We present *Annotation Cascading*: a framework describing how web content routes through a hierarchy of increasingly specialised language models. The framework is consistent with public engineering disclosures from major search operators: smaller, domain-specific models handle annotation wherever confidence permits, with LLM engagement reserved for cases where cheaper models cannot classify content confidently. Microsoft’s Bing engineering team has publicly documented their transition to tiered SLM/LLM architectures optimised for latency and cost (Microsoft Bing, 2024). Google Research has published work demonstrating that small, decomposition-based models achieve superior intent extraction compared to monolithic large models (Cohen et al., 2025).

The stakes of annotation quality are increasing. The discipline of optimising content for algorithmic systems has evolved from search engine optimisation (SEO), through Answer Engine Optimization (AEO, Barnard 2017) and AI Assistive Engine Optimization (AIEO, Barnard 2024a), to Assistive Agent Optimization (AAO, Barnard 2025c), which addresses AI that acts autonomously on behalf of users. We use AAO as a label for the broader class of autonomous decision systems in which retrieval and entity resolution errors can directly trigger incorrect actions. When an AI agent acts rather than recommends, annotation accuracy becomes the foundation on which autonomous action reliability depends.

This reframing has three consequences developed in this paper. First, Annotation Cascading (Section 4): content annotation is modelled as a hierarchical routing problem with four levels, where confident misannotation — defined as domain routing error compounded by control-plane failure — is identified as a structurally distinct failure mode. Second, topical authority as annotation routing efficiency (Section 5): we propose a formal definition with measurable dimensions, a cost decomposition model, and operationalisable metrics. Third, the Navigation Memory hypothesis (Section 6): we hypothesise that annotation context propagates between pages during a crawl session, and provide experimental designs with explicit threats to validity.

1.1 Scope and Epistemic Status

Annotation Cascading and Navigation Memory are conceptual frameworks grounded in public engineering disclosures and computational constraint analysis. We do not claim to describe the

internal implementation of any specific search engine or AI system. Rather, we propose a model that is *consistent with* observable system behaviour, published engineering decisions, and fundamental cost-performance trade-offs in web-scale classification. Throughout this paper, we provide testable predictions and measurement protocols, not confirmation of proprietary implementations. Where we describe mechanisms, we do so in the language of hypothesis and model, distinguishing between claims supported by direct evidence and those inferred from system constraints.

1.2 Annotation Scope: A Taxonomy

“Annotation” in web-scale systems encompasses several distinct tasks. To bound our claims, we distinguish four levels:

A1: Topical classification. Assigning topic or category labels to content (e.g., “health,” “cardiology,” “insurance”). This determines which specialist model family processes the content.

A2: Entity linking. Mapping entity mentions in content to Knowledge Graph entries. This establishes the entity-topic-URL associations central to our topical authority model.

A3: Attribute and relation extraction. Identifying properties of and relationships between entities (e.g., “Dr. Smith treats cardiac arrhythmia at Mayo Clinic”).

A4: Index feature annotation. Quality signals, schema interpretation, E-E-A-T evaluation, and other features used in downstream ranking and retrieval.

The Annotation Cascading framework primarily models **A1 (topical classification)** and **A2 (entity linking)** as the routing-dependent stages of the pipeline. A3 and A4 are downstream consequences: once content is routed to the correct specialist, attribute extraction and index feature computation proceed within that specialist’s domain. Errors at A1–A2 can propagate into A3–A4; we treat A3–A4 as downstream in this paper. This scoping allows falsifiable predictions about routing efficiency without requiring claims about the full annotation stack.

2. Related Work

The contributions in this paper intersect several established research areas. We situate each contribution against the relevant literature.

2.1 Multi-Stage Classification and Cascade Architectures

Hierarchical classification cascades have a long history in information retrieval and machine learning. Multi-stage pipelines that route documents through increasingly specialised classifiers are well established in text categorisation (Johnson & Zhang, 2017; Kim, 2014; Sun et al., 2019). Adaptive computation and early-exit architectures, where inference terminates at a cheaper

model when confidence is sufficient, have been explored extensively (Teerapittayanon et al., 2016; Bolukbasi et al., 2017). Mixture-of-experts architectures achieve efficiency by routing inputs to specialised sub-models (Shazeer et al., 2017; Fedus et al., 2022). At web scale, industrial retrieval systems employ multi-stage ranking and classification pipelines in which candidate generation, scoring, and re-ranking are performed by models of increasing complexity (Nogueira & Cho, 2019; Wang et al., 2011). Annotation Cascading draws on these traditions but applies them specifically to web-scale content annotation, identifying confident misannotation — routing to the wrong specialist domain with high confidence — as a structurally distinct failure mode not addressed in the early-exit literature, which focuses primarily on confidence thresholds for escalation within a single model family.

2.2 Entity Linking and Knowledge Graph Construction

The annotation routing model depends on entity-URL mappings as routing anchors, connecting to the entity linking literature (Bollacker et al., 2008; Vrandečić & Krötzsch, 2014). Modern entity linking systems face the same cost-accuracy trade-offs at scale (Sevgili et al., 2022; Wu et al., 2020), and production deployments increasingly rely on tiered approaches where cheap string-matching resolves unambiguous mentions and expensive neural models handle difficult cases. The `isReferencePage` attribute documented in the 2024 Google API leak (Jones, 2024) suggests that production systems may maintain explicit entity-URL designations, consistent with anchor-based entity resolution strategies in the knowledge graph literature.

2.3 Graph-Based Priors and Label Propagation

The Navigation Memory hypothesis is conceptually related to graph-based classification methods where labels propagate through document graphs (Zhu et al., 2003), and to contextual priors in web crawl processing where classification of a page is informed by its graph neighbourhood. The distinction between Carry Anchors and Break Anchors extends this literature by proposing that link topology functions as an annotation context management mechanism, not merely a relevance or authority signal.

2.4 Authority, Expertise, and Topical Clustering

Topical authority has been extensively discussed in the SEO practitioner literature (Gübür, 2020; Enge et al., 2015) and is implicitly referenced in Google’s E-E-A-T quality rater framework. However, no formal definition exists in the information retrieval literature. Related concepts include expertise retrieval (Balog et al., 2012), where the goal is to identify authoritative entities for a given topic, and topical clustering in web retrieval, where thematic coherence influences retrieval performance. Our contribution is to operationalise topical authority as annotation routing efficiency — a measurable property grounded in system constraints rather than an evaluative abstraction.

2.5 Retrieval-Augmented Generation and Annotation Quality

The increasing reliance of large language models on retrieval-augmented generation (Lewis et al., 2020) makes annotation quality directly consequential for downstream generation. Content that is poorly annotated may be retrieved in incorrect contexts, producing errors that propagate into AI-generated responses.

3. Terminology

In selecting terminology, we apply the principle that domain terms should be comprehensible to informed non-specialists without requiring a glossary definition. We use “Annotation Cascading” rather than “hierarchical model routing with fallback escalation,” “Navigation Memory” rather than “inter-page annotation context inheritance,” and “Assistive Agent Optimization” rather than “autonomous AI action reliability engineering” for this reason, while retaining technical precision in formal definitions below.

Annotation Cascading. The process by which web content routes through a hierarchy of increasingly specialised language models during classification. Each routing level either confirms the upstream assignment or overrides it. The large language model serves as the computational fallback: engaged when specialist models cannot achieve sufficient classification confidence.

Topical Authority. The measurable ease with which an algorithmic system can route content about a given entity-topic pair to the appropriate specialist annotation model.

Navigation Memory. The hypothesised persistence of annotation context between consecutive pages during a crawl session. When a crawler follows a link from page A to page B, the specialist model context established for page A may influence the initial routing decision for page B.

Assistive Agent Optimization (AAO). The discipline of optimising entity representation for AI systems that act autonomously on behalf of users, extending AEO (Barnard 2017) and AIEO (Barnard 2024a) to address the shift from AI that recommends to AI that acts (Barnard 2025c).

Specialist Language Model (SLM). A language model trained or fine-tuned for a specific domain, topic, or classification task. SLMs are computationally cheaper than general-purpose LLMs and produce higher accuracy within their domain of specialisation.

Confident Misannotation. A compound failure mode consisting of (1) a domain routing error, in which content is assigned to the wrong specialist model, and (2) a control-plane failure, in which the routing confidence exceeds the escalation threshold, preventing detection and correction. The combination produces systematic classification errors invisible to quality-control mechanisms that use confidence thresholds as their primary filter.

Entity Home. The canonical web property that anchors an entity’s identity across all knowledge representations (Barnard, 2015; 2022; 2025b).

4. Annotation Cascading: The Routing Hierarchy

4.1 The Four-Level Cascade

We propose that web content annotation (primarily tasks A1 and A2 as defined in Section 1.2) can be modelled as a four-level routing cascade. Each level corresponds to a routing decision that selects a specialist model family; the same family may then perform multiple annotation tasks (topical classification, entity linking) at that level. The levels represent successive *routing granularities*, not necessarily separate model stages in a fixed pipeline — implementations may collapse or parallelise levels depending on system architecture.

Level 1 — Site classification. The system identifies the domain’s primary topic and entity associations. For illustration, a URL on a health-focused domain (e.g., a medical publisher) may be initialised using prior signals — crawl history, Knowledge Graph entity associations, structured data — to route to health-domain specialist models before page content is analysed.

Level 2 — Category classification. Within the site’s domain, the system identifies the content category. A page in a /diseases-conditions/ subdirectory of a medical site routes to a conditions-classification specialist, narrowing from the broad health domain. Category signals include URL path structure, breadcrumb markup, and internal linking patterns.

Level 3 — Page classification. The system analyses the page’s primary content to determine topic, entities, and relationships. The specialist model assigned at Level 2 processes the page. If the content matches the expected category, annotation proceeds at low cost. If the content diverges, the system must reroute, incurring additional computational cost.

Level 4 — Chunk classification. Individual content segments within the page may require different specialist models. A medical article containing a biographical section, a technical section on treatment protocols, and a commercial section on pricing presents three distinct annotation contexts within a single URL. Chunking policies vary across implementations (DOM blocks, passages, semantic segments); the cascade model is agnostic to the specific chunking strategy.

4.2 The LLM as Computational Fallback

The critical reframing: we propose that the LLM is not the system’s preferred annotation mechanism but its most expensive option, engaged when the cascade cannot route content confidently. We define this operationally: *LLM fallback is invoked when cheaper routing and classification models cannot achieve target confidence within budget constraints (latency and*

cost). The measurable implication is that LLM-triggering content signals routing ambiguity or low model agreement, not superior understanding.

Three failure modes may trigger LLM escalation:

Routing ambiguity. The system cannot determine which specialist model should process the content. This occurs when content spans multiple domains without clear primary classification, when structured data signals conflict with content signals, or when the entity is unknown to the Knowledge Graph.

Confidence failure. The specialist model processes the content but produces low-confidence outputs, triggering verification by a more capable model. If the more capable model also produces low confidence, the LLM is engaged as the final arbiter.

Confident misannotation. The most damaging failure mode, defined as a compound of two failures. First, a *domain routing error*: the cascade assigns content to the wrong specialist model (e.g., a financial services page routed to a health-domain specialist because it discusses “health insurance plans”). Second, a *control-plane failure*: the routing confidence $c(U)$ exceeds the escalation threshold τ , so no escalation is triggered. The error propagates into downstream systems undetected. This is structurally distinct from miscalibration within a correctly routed specialist (Guo et al., 2017): the issue is not that the model’s confidence is poorly calibrated, but that the model is confidently operating in the wrong domain.

Detection of confident misannotation, by definition, cannot rely on confidence thresholds. We propose three detection signals: (1) contradiction by downstream entity resolution (entity links that conflict with the assigned topic domain), (2) disagreement between two orthogonal specialists applied to the same content (e.g., a topic classifier and an entity linker producing incompatible assignments), and (3) later reroute events in which the same content is reassigned on a subsequent crawl pass. We operationalise this as the **domain–entity incompatibility rate**: the fraction of pages on a domain where the topic model assigns domain T^d (the assigned topic-domain label) but the entity linker assigns entity types incompatible with T^d (e.g., financial service entities extracted from pages classified in the health domain). A non-trivial incompatibility rate on a domain would constitute evidence of systematic confident misannotation.

4.3 Implications for Content Architecture

If annotation is a routing problem, then content architecture functions as a routing signal. Every structural decision a publisher makes — URL hierarchy, internal linking, breadcrumb markup, Schema.org structured data, heading structure — either confirms or contradicts the routing decisions the system is attempting to make.

Content that is topically consistent within its URL path, structurally predictable in its page layout, and semantically aligned with its site’s established domain would, under this model, produce clean routing: the specialist model assigned at Level 1 or Level 2 handles the entire annotation at minimal cost. Content that is topically inconsistent, structurally unpredictable, or semantically divergent forces rerouting, escalation, and potentially LLM fallback.

This provides a mechanistic explanation, consistent with the cascade model, for a widely reported practitioner observation: topically coherent sites consistently outperform topically diverse sites of equivalent quality (Gübür, 2020; Enge et al., 2015). We hypothesise that annotation routing efficiency contributes to this effect, though we acknowledge that other factors (crawl efficiency, link equity concentration, user engagement signals) likely also play a role.

5. Topical Authority as Annotation Routing Efficiency

5.1 The Problem of Definition

“Topical authority” is among the most discussed and least defined concepts in the search engine optimisation literature. Practitioners use the term extensively (Gübür, 2020; Barnard, 2015; Enge et al., 2015). Google’s quality rater guidelines reference authority as a component of E-E-A-T. Yet no formal definition exists in the information retrieval literature that would allow measurement, comparison, or falsification. We propose one.

5.2 Definition

Topical authority is the annotation routing efficiency of an entity-topic pair: the measurable ease with which an algorithmic system can route content about entity E on topic T to the appropriate specialist annotation model. High topical authority means the system tends to assign new content to a specialist model at earlier cascade levels (Level 1 or Level 2), with high confidence and low computational cost. Low topical authority means the system must escalate through the cascade.

5.3 Formal Model

Let E denote an entity, T a topic, and U a URL. Define association weight $w(E, T, U)$ as a composite signal derived from Knowledge Graph links, structured data, anchor text context, and content embeddings linking entity E to topic T via URL U .

For a given URL U , the annotation system makes a routing decision $r(U) \in \{m_1, m_2, \dots, m_k\}$, selecting from k available specialist models, with associated confidence $c(U)$. Escalation is triggered when $c(U) < \tau$ (the confidence threshold) or when model disagreement exceeds δ (a divergence threshold between orthogonal annotation signals).

The total annotation cost for a URL decomposes across cascade levels. Let L denote the number of cascade levels executed for URL U . The annotation cost is:

$$C(U) = \sum_{\ell} \mathbb{1}[\text{level } \ell \text{ executed}] \cdot \text{cost}(m_{\ell})$$

where m_{ℓ} is the model selected at level ℓ and $\text{cost}(m_{\ell})$ captures the computational expense (inference time, memory, API cost) of that model. For content with high topical authority, the cascade resolves at Level 1 or Level 2: only cheap specialist models are invoked, and $C(U)$ is low. For content with low topical authority, escalation through Level 3 or to the LLM fallback at Level 4 produces high $C(U)$.

Annotation routing efficiency for an entity-topic pair (E, T) is then the expected annotation cost across the entity’s URL set, given a target accuracy threshold A . Let $U_{e,T}$ be the set of URLs associated with (E, T) . Routing efficiency is $\mathbb{E}[C(U)]$ for $U \sim U_{e,T}$ at target accuracy A : lower expected cost at equivalent accuracy indicates higher topical authority. In black-box settings where $\text{cost}(m_{\ell})$ cannot be observed directly, cost can be approximated by proxy indicators of escalation: instability of entity extraction across crawl passes, variability in topic classification over time, or the presence of reclassification events — each indicating that cheaper models failed to resolve the annotation confidently.

5.4 The Three Dimensions

The three dimensions of topical authority map to measurable quantities within this cost framework:

Closeness = intra-site topical entropy or embedding dispersion across the entity’s URLs. An entity whose content clusters tightly around a topic has low dispersion and high closeness, reducing routing ambiguity at Level 2. Formally: lower entropy in the distribution of topic assignments across the entity’s URL set corresponds to higher closeness. High closeness reduces the probability of escalation, lowering expected $C(U)$.

Strength = corroboration count from independent sources: the number of high-authority third-party references confirming the entity-topic association, weighted by source authority. Strength reduces confidence failure at Level 3 by providing external evidence that supports the routing decision, increasing $c(U)$ and reducing the probability that $c(U) < \tau$.

Longevity = time-weighted stability of $w(E, T, U)$: the duration over which the entity has maintained consistent association weights with the topic. Longer temporal stability provides stronger evidence that the routing decision is durable. A system that has successfully routed content from entity E on topic T for ten years can assign specialist models with higher prior confidence than for a six-month publishing history.

This formalisation does not require access to proprietary system internals. Each dimension can be approximated using publicly observable signals: content analysis for closeness, backlink and citation analysis for strength, and temporal web archive data for longevity.

5.5 Empirical Consistency: *isReferencePage*

Documentation from the May 2024 Google Search API leak includes an *isReferencePage* boolean, described as indicating that the entity matches the full URL of the document (Jones, 2024; Fishkin, 2024). This attribute is *consistent with* a system that maintains explicit entity-URL mappings used as anchors for downstream entity handling (Barnard, 2024b). We note that leaked documentation does not constitute confirmed production usage, and alternative interpretations are possible. However, the existence of such a field is consistent with the annotation routing model: a URL designated as reference for an entity could serve as the canonical source for that entity’s topic associations, potentially enabling high-confidence routing at Level 1.

5.6 Relationship to G    r’s Semantic SEO Framework

Koray Tu  berk G    r’s work on topical authority and semantic SEO (G    r, 2020) established the practitioner framework for understanding how topic clusters and entity-topic associations influence search visibility. His central insight — that search engines evaluate topical comprehensiveness rather than individual page quality — is consistent with the annotation routing model and can be understood as an empirical observation of the routing efficiency mechanism proposed here. Where G    r’s framework describes what to build, the annotation routing model proposes an explanation for why it works.

6. Navigation Memory Hypothesis: Inter-Page Context Propagation

6.1 The Hypothesis

We hypothesise that when a crawler processes multiple pages on a site in sequence, the annotation context does not fully reset between pages. The specialist model loaded for page A — its topic classification, entity associations, and confidence level — may carry forward as a prior when the crawler follows an internal link to page B. This is not a claim about persistent state in the software engineering sense. It is a hypothesis about computational efficiency: if the system has already loaded and configured a specialist model for a topic, it *would be cheaper* to apply that model to the next page than to restart the routing cascade from Level 1.

6.2 Supporting Evidence

The evidence for Navigation Memory is inferential rather than directly confirmed, but converges from multiple directions.

Microsoft’s public documentation of Bing’s tiered SLM/LLM architecture emphasises computational efficiency as the primary design constraint (Microsoft Bing, 2024). A system optimised for efficiency would have reason not to discard loaded model context between pages when the next page is likely to require the same specialist model, particularly when the crawler follows internal links within a topically coherent site.

Empirical observations from the SEO practitioner community consistently report that internal linking structure affects not just discovery and crawl priority but classification quality: pages linked from topically relevant parents appear to receive more consistent classification than orphan pages or pages linked from unrelated contexts. While multiple mechanisms could contribute to this effect, some form of inter-page context propagation is one plausible explanation.

6.3 Implications: Carry Anchors and Break Anchors

If Navigation Memory operates as hypothesised, then internal links function as context propagation channels in addition to their roles in navigation and authority distribution. We propose two categories:

Carry Anchors. Internal links that maintain the current annotation context. A link from a cardiology article to another cardiology article carries the specialist model context forward. The anchor text confirms the topic. The destination page benefits from accumulated routing confidence.

Break Anchors. Internal links that force a context reset. A link from a cardiology article to a billing FAQ breaks the specialist model context. The system must restart routing from Level 1 or Level 2 for the destination page.

6.4 Proposed Experimental Designs

To move Navigation Memory from hypothesis to empirically tested mechanism, we propose two experimental designs:

Experiment 1: A/B site graph. Create two identical sets of pages on controlled test domains. In condition A, pages are linked in topically coherent chains (cardiology → cardiology → cardiology). In condition B, the same pages are linked through topically unrelated intermediaries (cardiology → billing → cardiology). Use server-side crawl logs to confirm crawl ordering, and measure classification outcomes via proxies defined below.

Experiment 2: Convergent path analysis. Identify a single target page reachable via two different internal link paths — one topically coherent, one topically divergent. Use sitemap directives and server log analysis to establish which path was crawled. Measure whether the page’s classification or feature eligibility differs depending on crawl path.

6.5 Proxy Metrics and Threats to Validity

We propose three tiers of proxy metrics, ordered by proximity to the underlying annotation mechanism:

Tier 1 (strongest): Entity extraction consistency. Stability of extracted entities in publicly observable features (structured snippet elements, entity cards, consistent entity disambiguation in surfaced annotations, and, where applicable, Knowledge Panel attributes) for identical content reached via different link paths. This is the closest observable proxy to annotation-level classification.

Tier 2 (moderate): Independent classifier agreement. Apply multiple open-source topic classifiers and entity linkers to the same content, varying only the contextual framing (preceding page content as input prefix). Measure whether classifier outputs shift as a function of contextual input, simulating the Navigation Memory mechanism outside proprietary systems.

Tier 3 (weakest): Search Console observables. Indexing patterns, rich result eligibility, snippet consistency, and canonical topic classification as reflected in search features. These are influenced by many non-annotation factors.

Threats to validity. Several confounds must be controlled. Search Console observables (Tier 3) are influenced by ranking algorithms, A/B testing, personalisation, and UI experiments independent of annotation. Crawl ordering cannot be fully controlled on production search engines; server logs provide only partial evidence of crawl sequence. Entity extraction consistency (Tier 1) may reflect KG-level caching rather than annotation-level propagation. The independent classifier approach (Tier 2) simulates the hypothesised mechanism but cannot confirm that production systems implement it. Additionally, the observed effects attributed to Navigation Memory could arise from alternative mechanisms: graph-neighbourhood priors in the knowledge graph, training-set biases that correlate with link structure, or templated page similarity within topically coherent sites. These limitations motivate a multi-tier proxy strategy: convergent evidence across tiers is more compelling than any single proxy.

7. Predictions, Proxies, and Expected Directions

Table 1 maps each framework component to its testable predictions, measurable proxies, and expected directional effects.

Table 1. *Predictions-to-proxies mapping for the unified annotation model.*

Component	Prediction	Measurable Proxy	Expected Direction
Annotation Cascading	Higher topical coherence reduces escalation	Escalation rate (proxy): proportion of pages exhibiting instability markers consistent with higher-tier processing	Coherent sites: lower escalation, lower reroute rate
Annotation Cascading	Confident misannotation produces invisible errors	Entity-topic contradiction rate; downstream reroute events	Mixed-domain sites: higher contradiction rate
Topical Authority (Closeness)	Tighter topical clustering reduces routing ambiguity	Intra-site topic entropy; embedding dispersion	Lower entropy → lower escalation cost
Topical Authority (Strength)	More corroboration increases routing confidence	Third-party citation count; KG linkage density	Higher corroboration → higher $c(U)$, fewer confidence failures
Topical Authority (Longevity)	Temporal stability increases prior confidence	Time-weighted consistency of $w(E,T,U)$ via web archives	Longer history → earlier cascade resolution
Navigation Memory	Carry Anchors preserve context; Break Anchors reset it	Entity extraction consistency across link paths (Tier 1); classifier agreement shift (Tier 2)	Carry Anchor paths: higher classification stability, fewer reroutes

Note: “Instability markers” refers to observable signals consistent with escalation beyond specialist-model processing: variability in entity extraction across crawl passes, inconsistent topic classification over time, reclassification events, or entity-topic contradictions. These are proxies for escalation likelihood rather than direct cost measurements, but are sufficient to compare relative routing efficiency across conditions.

8. The Unified Annotation Model

Annotation Cascading, topical authority as routing efficiency, and Navigation Memory form a unified model of how algorithmic systems may classify web content at scale. The cascade determines how content routes through the model hierarchy. Topical authority determines how efficiently and confidently that routing occurs. Navigation Memory describes how routing context may propagate between pages during a crawl session.

Together, they offer a mechanistic explanation for why topically coherent, well-structured, consistently maintained content from established entities outperforms fragmented, inconsistent, or topically scattered content, even when the scattered content may be individually superior in

quality. The proposed mechanism is not abstract “relevance” but measurable annotation efficiency.

9. Limitations and Future Work

The Navigation Memory hypothesis (Section 6) is supported by inferential evidence and computational efficiency arguments but lacks direct confirmation. The experimental designs and proxy validation strategy proposed in Sections 6.4–6.5 would provide initial empirical validation; we intend to conduct these as a follow-up study.

The topical authority formalisation (Section 5) provides a cost model with operationalisable metrics, but requires validation against large-scale retrieval datasets to confirm that annotation routing efficiency correlates with downstream retrieval performance as predicted.

The annotation taxonomy (Section 1.2) scopes our claims to A1 (topical classification) and A2 (entity linking). Whether the cascade model extends to A3 (attribute extraction) and A4 (index features) as independent routing stages, or whether these are always downstream of A1–A2 routing, remains an open question.

The relationship between annotation quality and autonomous agent reliability (the AAO context) is asserted based on the logical chain from annotation accuracy through retrieval quality to action correctness, but is not empirically tested. As autonomous AI agents become more prevalent, measuring this causal chain will be critical.

Future work should address the temporal dynamics of annotation routing: how quickly a system updates its routing decisions when an entity’s topical focus shifts, and whether Navigation Memory effects decay between crawl sessions.

This paper describes a conceptual framework and does not disclose implementation details intended for patent protection.

10. Conclusion

We have proposed that annotation at web scale is best understood as a routing problem rather than a classification problem. A plausible design objective under cost constraints is to route every page to the cheapest model that can classify it with sufficient confidence. The LLM, under this model, is the computational fallback rather than the target.

Topical authority, reframed as the efficiency of that routing, becomes a measurable property with three operationalisable dimensions and a formal cost model. Navigation Memory, if confirmed, describes the persistence of routing context between pages, with internal linking functioning as context management. Confident misannotation — domain routing error compounded by control-plane failure — is identified as the structurally most damaging failure mode.

Content architecture, under this framework, is annotation architecture. Internal linking is context management. Site structure is a routing signal. And as AI systems evolve from recommending to acting, the quality of annotation at the foundation determines the reliability of autonomous action at the surface.

Use of Artificial Intelligence

The author used Claude (Anthropic) as a drafting and structural editing assistant during the preparation of this manuscript. All substantive claims, the conceptual framework, the formal model, and the interpretive analysis are the author's original work. Claude was used for prose refinement, structural organisation, and consistency checking. The author takes full responsibility for all content.

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